VIRGINIA TECH™

Animated Stimuli for *Aedes aegypti*

NICHOLAS GARDELLA, FALL 2020

Overview

01 Experiments

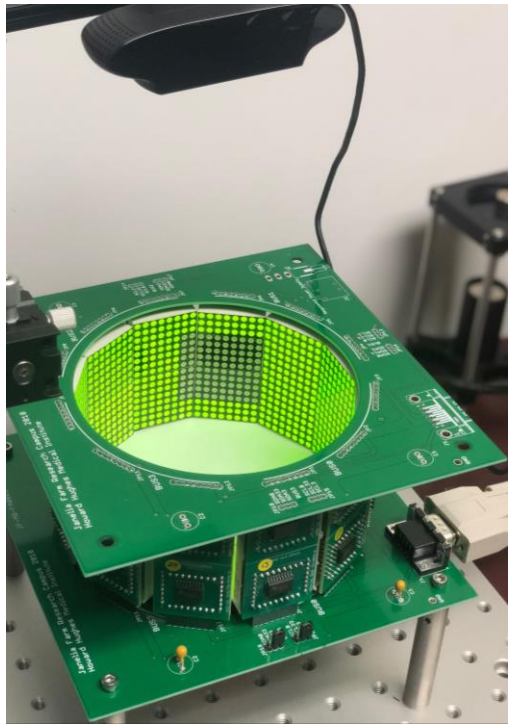
02 Technical Intro

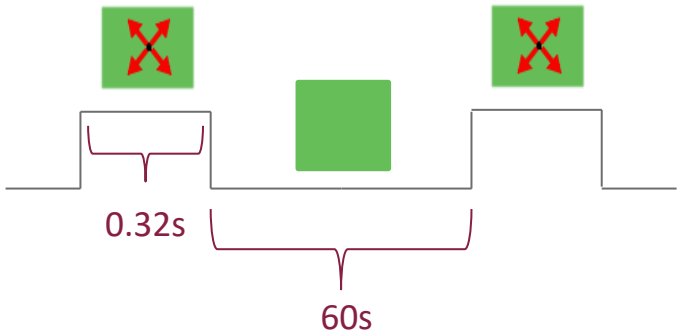
03 Strategy

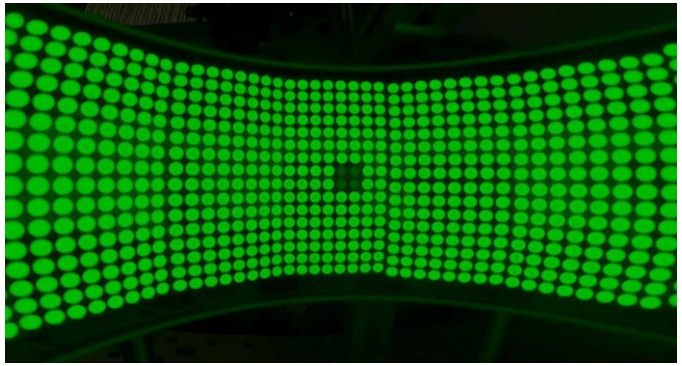
04 Products

01

EXPERIMENTS

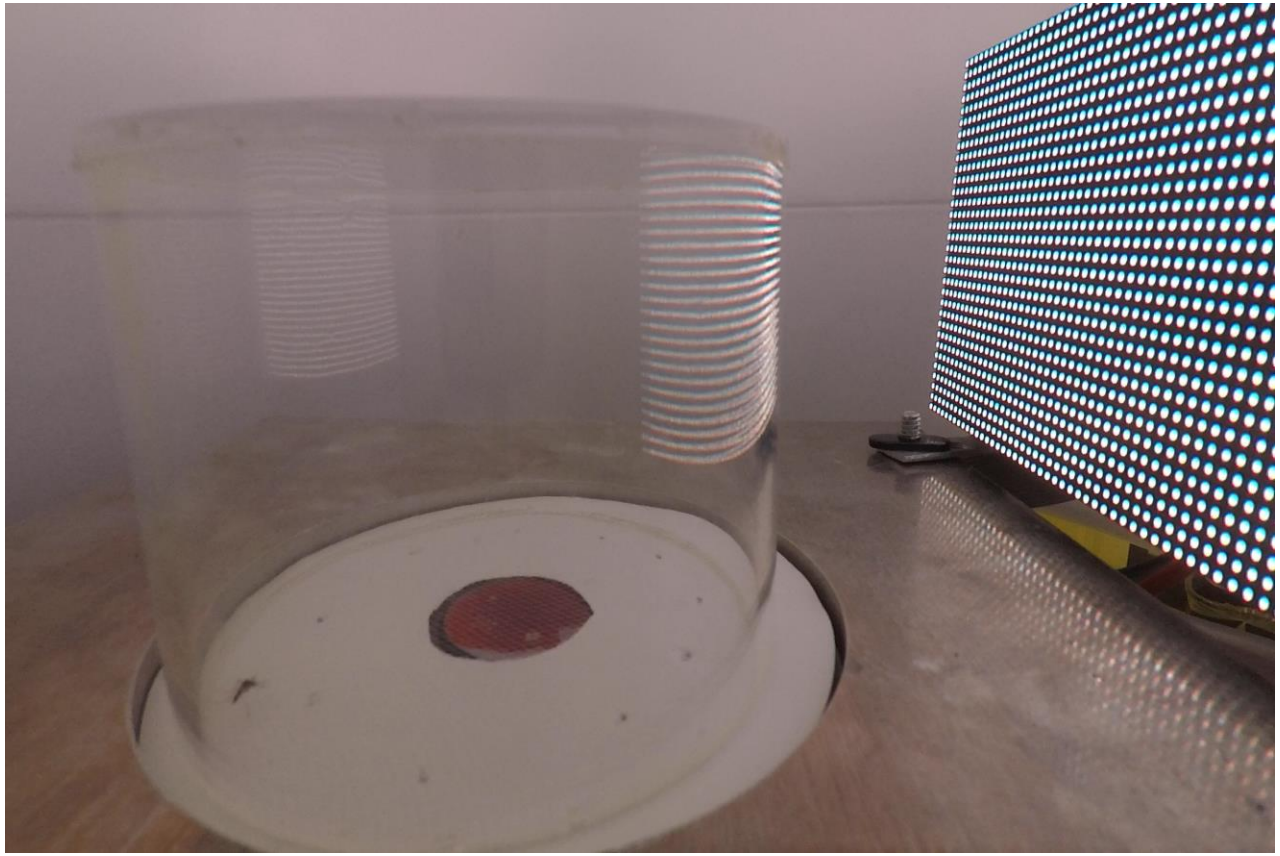






1. Escape Behavior

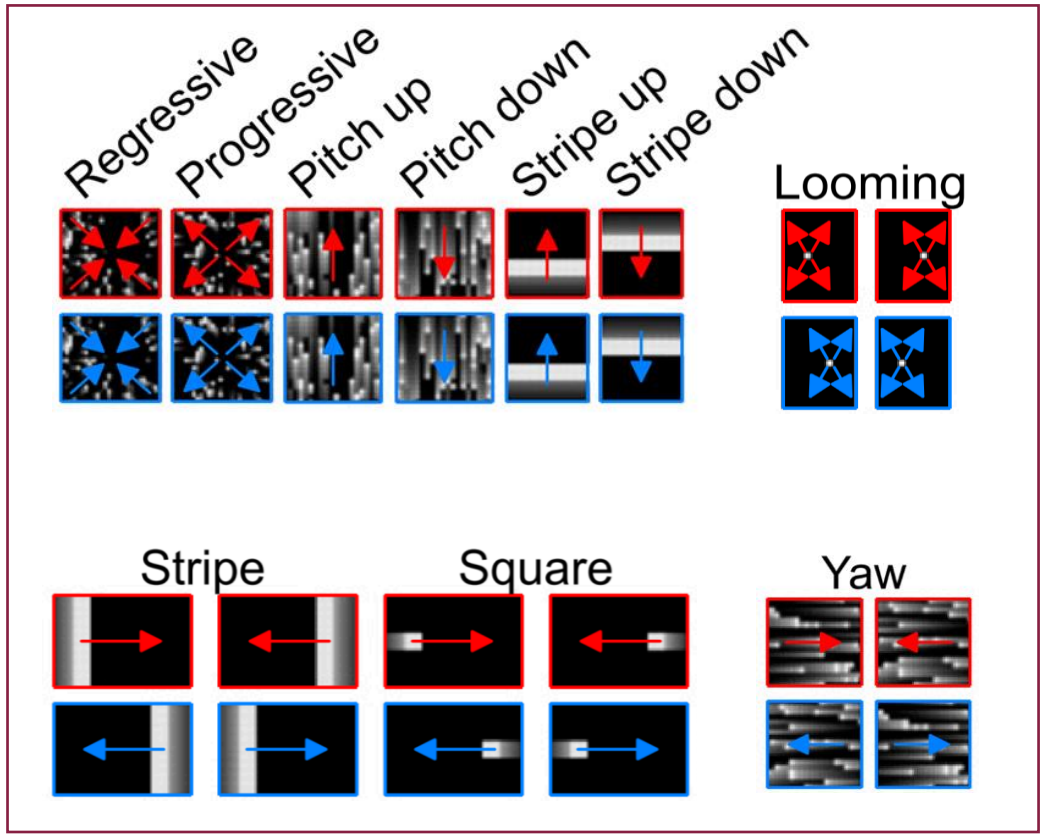
☒ Small arena with a looming stimulus in context of free flight



1. Escape Behavior

☒ Small arena with a looming stimulus in context of free flight

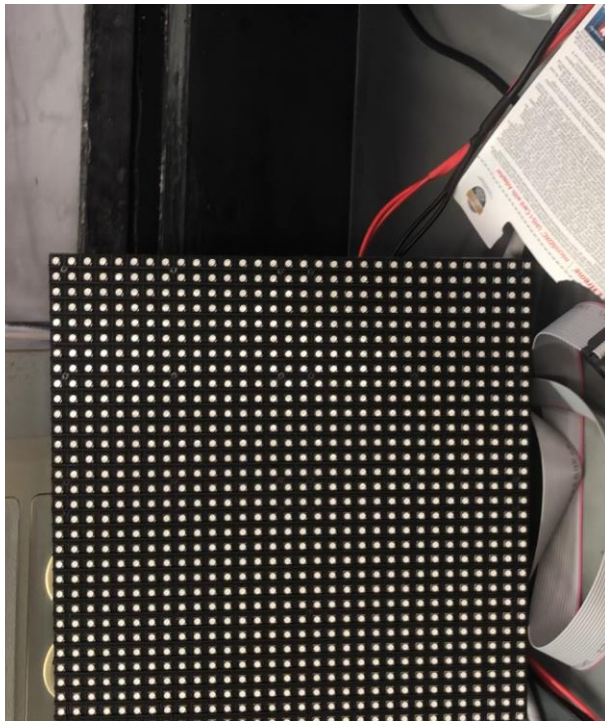
☐ Larger arena with a looming stimulus in context of blood feeding



P. T. Weir and M. H. Dickinson, "Functional divisions for visual processing in the central brain of flying Drosophila," *Proceedings of the National Academy of Sciences*, vol. 112, no. 40, p. E5524, 2015.

2. Visual Processing

☒ Electrophysiological recordings on genus *Drosophila*



2. Visual Processing

☒ Electrophysiological recordings on genus *Drosophila*

☐ Electrophysiological recordings on genus *Aedes*

02

TECHNICAL INTRO

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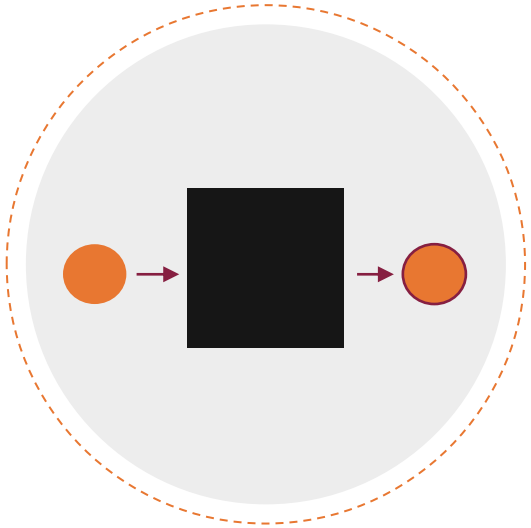
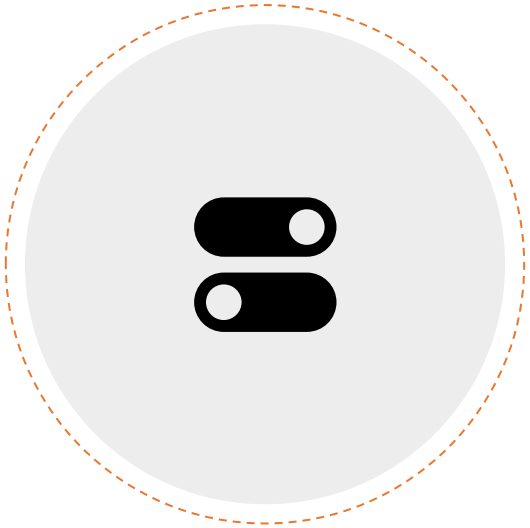


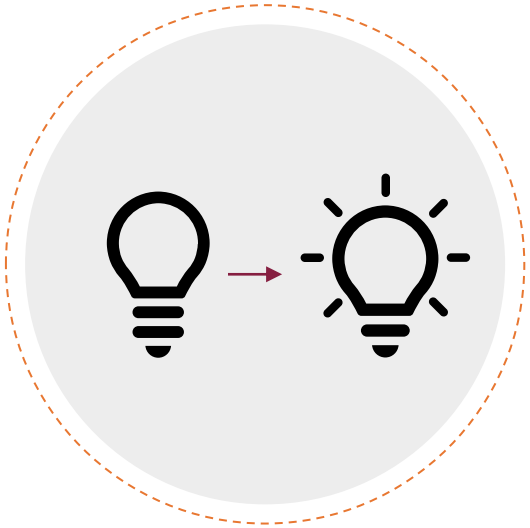
ABSTRACTION

"black box" theory



STATE

~~c = 299,792,458 m/s~~
v = 3.5 m/s North

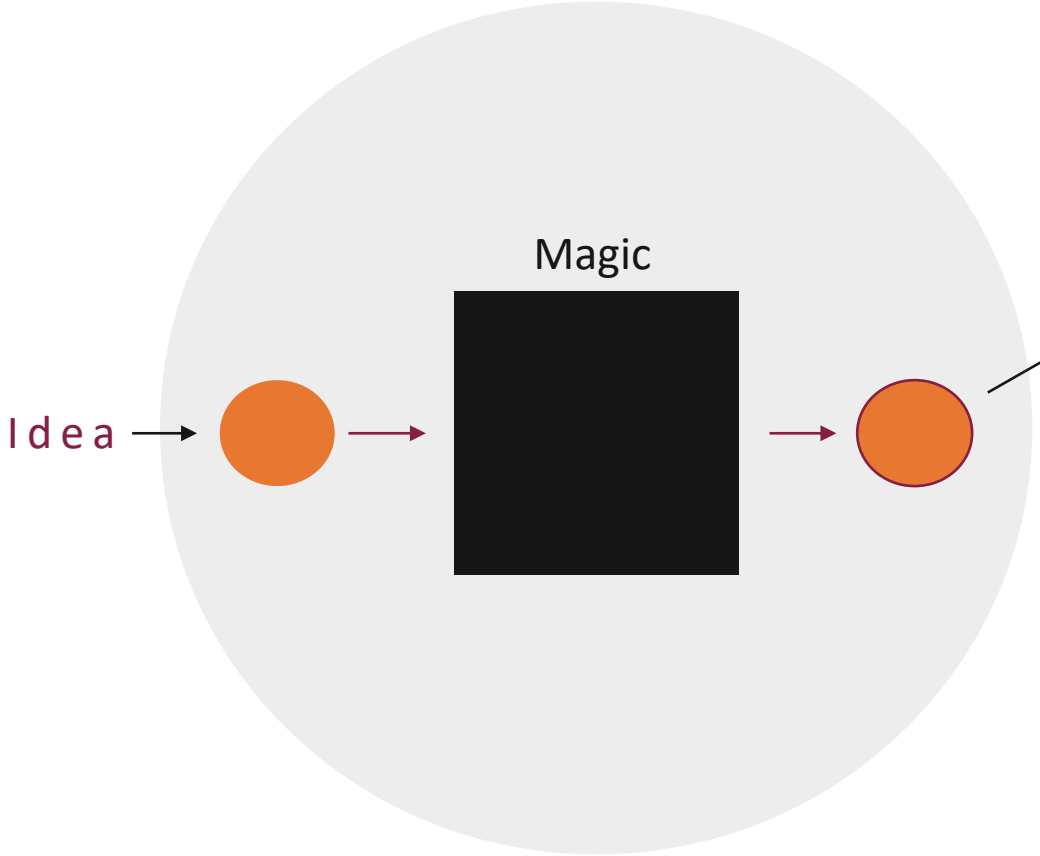


RENDERING

publishing changes
in state

03

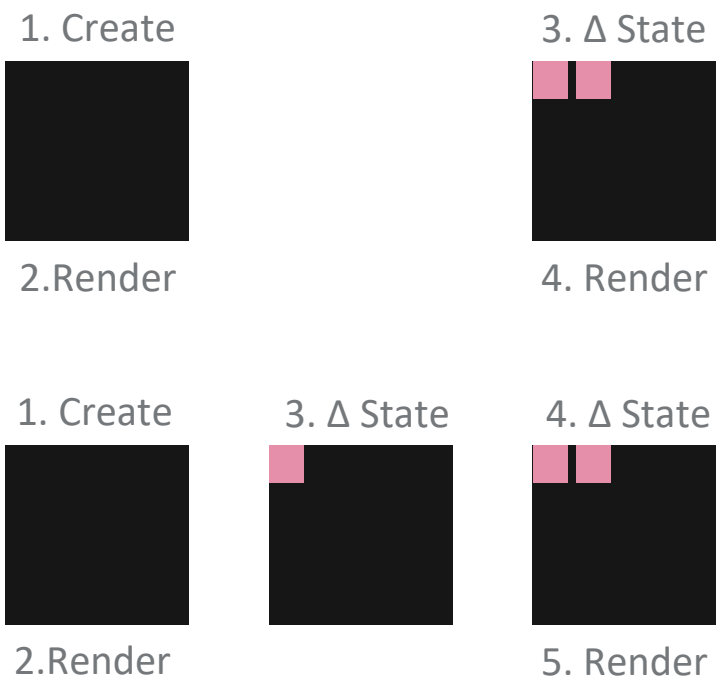
STRATEGY



Experimental Design

- Raspberry Pi OS
- hzeller/rpi-rgb-led-matrix
- Python Interpreter
- LED Display

[redacted]



Making Animations

1. Make a frame
2. Show the frame
3. Make changes
4. Show the frame
5. Repeat

Boring_Code.py

```
import NumPy as np
import random
import PIL as Image
import time
...
matrix = RGBMatrix()
height = self.matrix.height
width = self.matrix.height
...
def run(self):
        array = np.ones((width, height, 3), np.uint8)   # unABSTRACTION of the canvas
                                                        # to be conceptually a 3D array of #s

        img = Image.fromarray(array)                    # unABSTRACTION of the canvas
                                                        # to be conceptually an image
        canvas = matrix.CreateFrameCanvas()
        canvas.SetImage(img)                            # ABSTRACTION of the STATE of the display
                                                        # to be conceptually a canvas

        canvas = matrix.SwapOnVSync(canvas)             # RENDER changes in STATE

        time.sleep(0.1)                                 # pause before the next frame
```

Coding Animations

1. Import blackboxes
2. Understand interfaces
3. Conceptualize in state
4. Make it render-able
5. Render changes

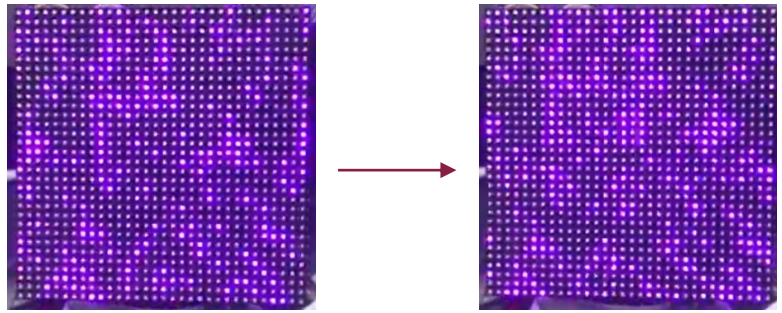
Desired Outputs

- Pitch/Yaw
- Looming Square/Circle
- Vertical/Horizontal Stripe
- Vertical/Horizontal Square
- Progression/Regression

```
progression(certain color, certain time, certain pause) {

        make a black screen and render it

        loop (repeat until 30% of the display has pixels turned on)
        {
                use random to get random numbers
                set that pixel to a certain color in the array
                split the array into quarters
                render it
        }

        loop (repeat until me when a certain time has elapsed)
        {
                roll each quarter away from the center
                render
                certain pause
        }

}
```

Example: Progression

- Change color
- Change speed
- Change density



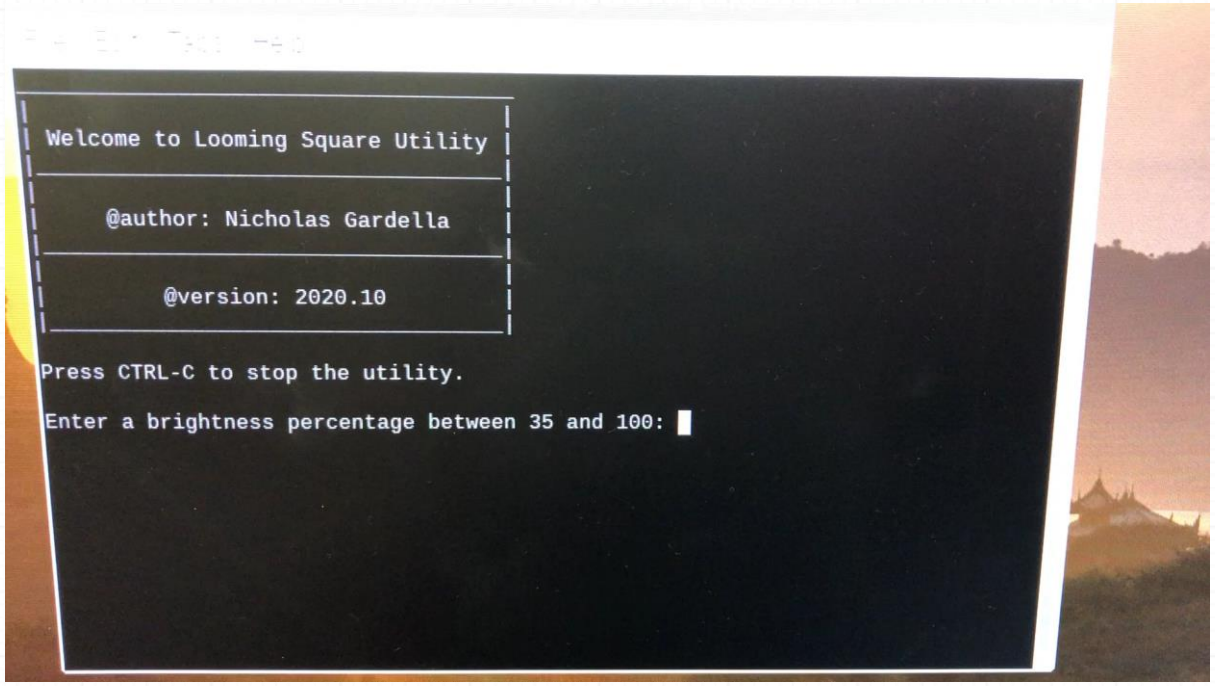
Example Deliverable

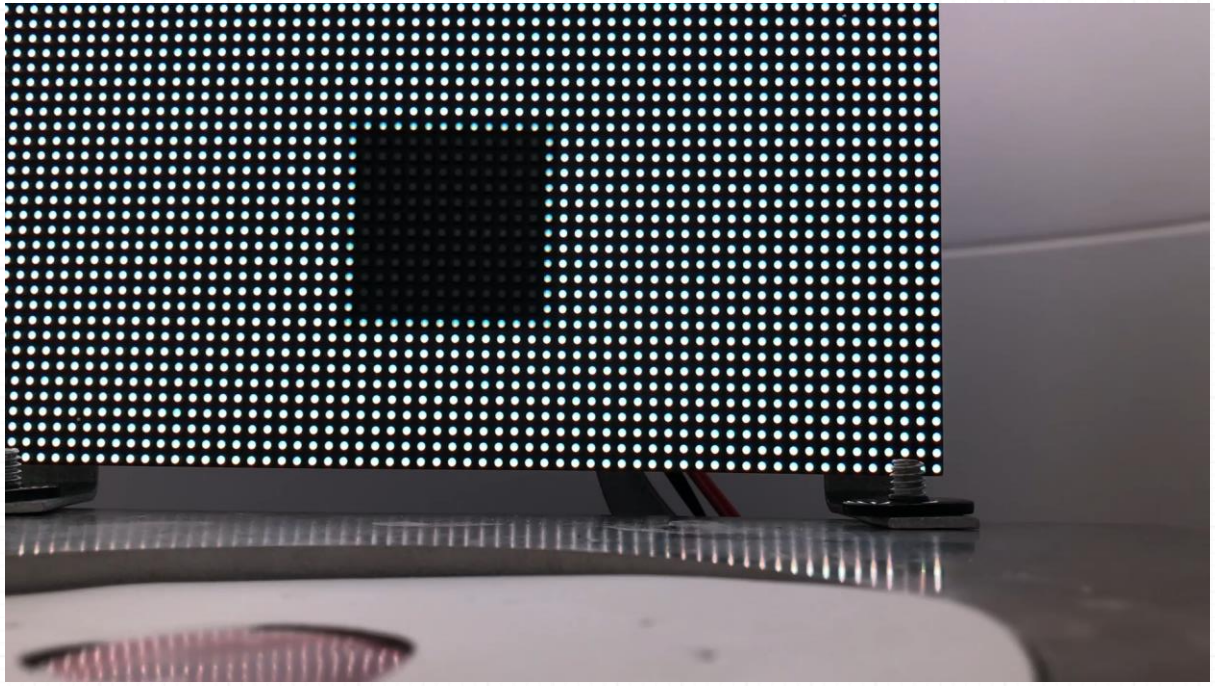
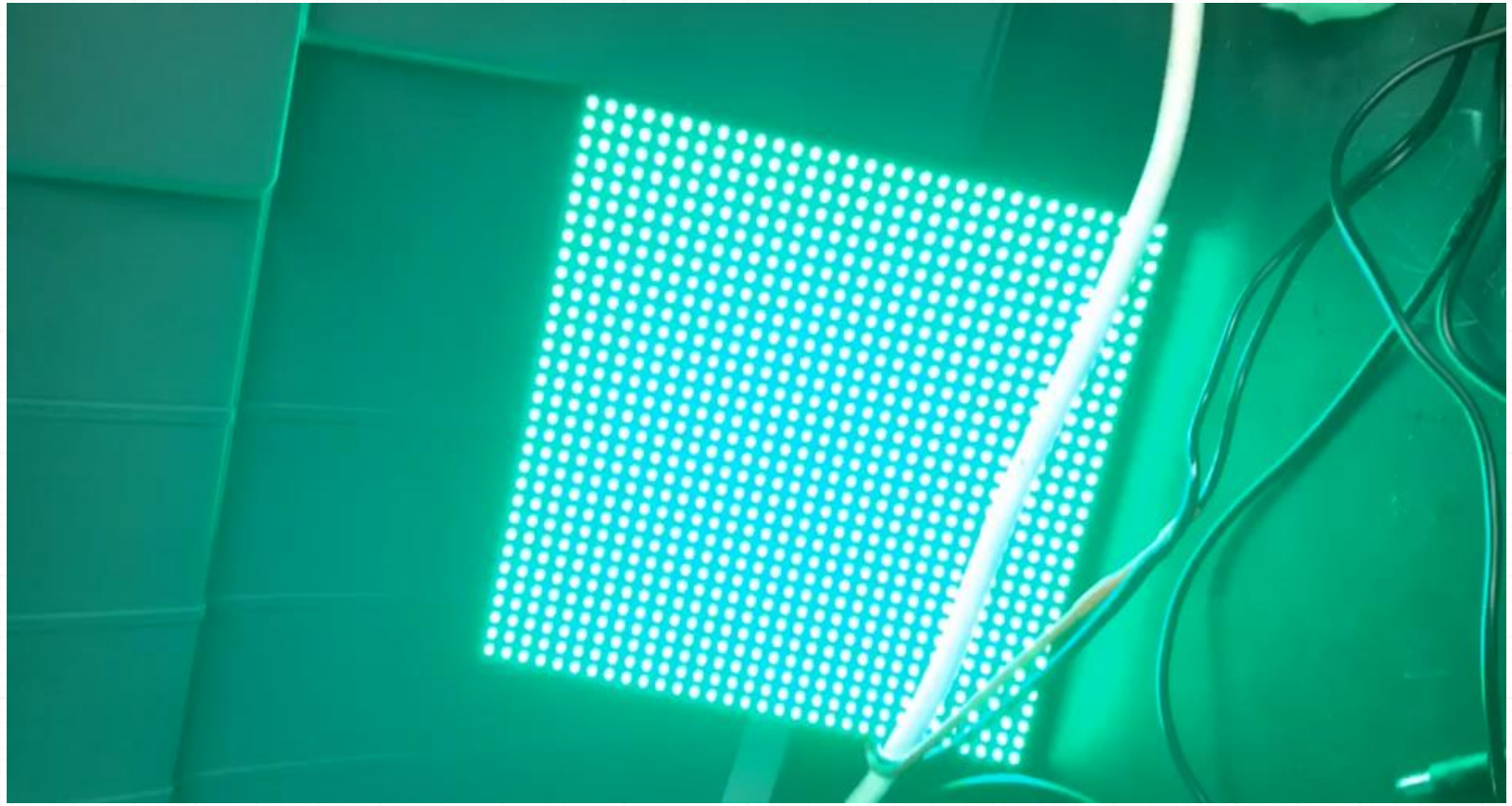
runProgressionAnimation(color, speed_multiplier, density)

04

PRODUCTS

Exp. 1 Final Product





Exp. 2 Final Product